

Preparation Quiz for the Midterm

Robotics and Automation

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Instructions: This quiz is intended as preparation for the upcoming midterm exam. Answer all questions and justify your reasoning when necessary.

Part I — Conceptual Questions

In robotics, many mechanical systems such as **robot joints, manipulators, and drones** can often be approximated by first- and second-order dynamical models.

1. What information do the **eigenvalues of a system matrix** provide about the behavior of a dynamical system used to model a robot?

2. Consider the first-order system

$$\dot{x} = -ax$$

with $a > 0$.

This type of model can represent the **velocity dynamics of a robot joint actuator**.

Is the system **stable or unstable**? Briefly explain your answer.

3. What is the role of the **derivative term (D)** in a PID controller when controlling a robotic system?

- a) Reduce steady-state error
- b) Increase system speed
- c) Add damping to the system
- d) Eliminate disturbances

4. Which term in a **PID controller** is primarily responsible for eliminating **steady-state error caused by constant disturbances**, such as friction or external forces acting on a robot?

- a) Proportional

- b) Integral
- c) Derivative
- d) Feedforward

Part II — Simulation-Based Question

In robotics, second-order models frequently describe the motion of **robot joints, manipulators, or mobile robots**.

Consider the system

$$\ddot{x} + 4\dot{x} + 5x = u$$

where x may represent the **position of a robotic joint**.

5. Use the **simulation code provided during the course** (the same Python code used in class) to help you analyze the system.

Answer the following:

- a) Is the system **stable** when $u = 0$?
- b) Describe how the state $x(t)$ behaves over time (for example: oscillatory decay, exponential decay, divergence, etc.).
- c) Explain how the **eigenvalues of the system** relate to the observed behavior.

Part III — State-Space Representation

The dynamics of many robotic systems (such as **robot arms or drones**) can be written in **state-space form**.

Consider the system

$$\ddot{x} + 4\dot{x} + 5x = u$$

where x represents the **position of a robot joint**.

6. Write the system in **state-space form**

$$\dot{x} = Ax + Bu$$

Define matrices A and B .

7. Compute the **eigenvalues of matrix A** and explain what they indicate about the **stability of the robotic system**.

Part IV — Control Design

Consider the simplified dynamics of a **robot joint or drone translational motion**

$$\ddot{x} = u$$

where x represents the **position of the robot**.

The system is controlled using a **PD controller**

$$u = -k_p x - k_d \dot{x}$$

8. Substitute the control law into the system and derive the **closed-loop differential equation**.
9. Explain how the parameters k_p and k_d influence the behavior of the robotic system (speed of response, oscillations, damping).
10. Suppose the robot must track a desired trajectory $x_d(t)$.

A control law with **feedforward** can be written as

$$u = -k_p(x - x_d) - k_d(\dot{x} - \dot{x}_d) + \ddot{x}_d$$

Explain:

- a) Why the term $\ddot{x}_d$ is included in the control law.
- b) What advantage feedforward provides compared with using only PD control when controlling a robot trajectory.